Problem 1 (Z79810E7)

Let C be a point on circle (AB) centered at M such that CY is tangent to the circle. Note that MCY, MXC, YXC are similar right triangles.

By similar triangles,

$$MA^2 = MC^2 = MX \cdot MY$$

By Power of a Point and similar triangles,

$$YA \cdot YB = YD^2 = YX \cdot YM$$

Problem 4 (22PAGMO3)

Note that $\angle PP_1Q = \angle PQ_1Q$ as they are equal to $90 - \angle BAP = 90 - \angle CAQ$ respectively. Therefore, PP_1Q_1Q is cyclic. Applying Brocard on it, we have AB is the polar of $PQ_1 \cap P_1Q$. However, $QP_1 \perp AB$ meaning the center of (PP_1Q_1Q) lies on P_1Q. We then have $\angle P_1PQ = 90$ meaning $\angle P_1Q_1Q = 90$ so $\angle BQ_1P = \angle BAP$. Therefore, AQ_1PB is cyclic. Since M lies on both angle bisectors M lies on the circle as well.

4 clubs; 1 hour

Problem 5 (12ESLG3)

Lemma (incircle polars): If $K = EF \cap BC$, then AD is the polar of K with respect to the incircle.

Proof: the pole of EF is A meaning the polar of K passes A. KD is tangent to the incircle so D lies on the polar as well. Thus, AD is the polar as desired.

Using the lemma, we get that in the problem, $IP \cap BC = EF \cap BC = K$. By lemma 9.11 from EGMO [Can I just cite it like so if I remember the lemma number but not the name in contest?], since AD, BE, CF are concurrent, we know that $(B, C; K, D) = -1$. Therefore, by another known lemma [and would it be alright if I only remember the statement and write "known lemma"] from EGMO, since $KP \perp AD$, we know $\angle BPD = \angle CPD$.

Problem 8 (21PAGMO6)

Claim: $A_1B_1 \cap B_2A_2$ is the midpoint of A_1B_1; $A_1C_1 \cap C_2A_2$ is the midpoint of A_1C_1.

Proof: Since $BI \perp BI_A$ and $\angle CBI = \angle IBA$, $(A, D; I, I_A) = -1$. Additionally, since $C_1A_1 \perp BI_A \perp BI$, we know that $BI \parallel C_1A_1$. Projecting onto C_1A_1, $(A, D; I, I_A) \overset{B}{\doteq} (C_1, A_1; Y, P_\infty) = -1$ meaning Y is the midpoint as desired. Same thing with A_1C_1 side.

Therefore, by a homothety at A_1 with scale 2, $AC_1 = AB_1 \Leftrightarrow MY = MX$.

10 clubs; 2.5 hours

Problem 9 (19ESLG1)

Let K be the point on (ABC) other than B such that $BK \parallel FE$. Let ray KE meets (ABC) again at X, we have X lies on $(AFHE)$ since $\angle AFE = \angle ABK = \angle AXK$. By radical axis on $(ARFE), (BCEF), (ABC)$, we know AR, EF, BC are concurrent. Call this point Y. Let $AP \cap BC = D$. By lemma 9.11, $(T, D; B, C) = -1$. Projecting with perspectivity at A onto (ABC), $(X, P; B, C) = -1$. Using perspectivity at E, where BE intersects (ABV) again at L, $(K, Q; L, A) = -1$. Finally, using perspectivity at B, $(P_{\infty \text{ on } EF}, M; E, F) = -1$ as desired.

Problem 11 (Z8AF8A34)

Let D, E be points on the paper such that $DE \cap AC = B$. Let F be a point on segment AB.

We draw $DF \cap AE = G$ and $EF \cap AD = H$. Then draw $BH \cap CD = I; BG \cap EC = J$. We prove that $EI \cap DJ$ lies on $\overline{ABC}$.

Note that the diagram is completely projective, so we send points A, E, C, D to a square $AECD$. The diagram is symmetric about AC meaning $EI \cap DJ$ lie on AC as desired.

Using this construction, note that as F approaches A on segment AB, G, H approach A on segments AE, AD. Therefore, I, J approach C on rays EC, DC from the other side of C as E, D respectively. Therefore, we may ensure that the distance from I, J to line BC is closer than that of E, D meaning we may ensure that $EI \cap DJ$ lies on the other side of C as A, B. We draw a ray at C through $EI \cap DJ$ to finish.

18 clubs; 4 hours

Problem 12 (02IRN)

Claim: If $MN \cap BC = K$, $(K, A'; B, C) = -1$.

The diagram is completely projective. We keep the incircle fixed and send $MN \cap PA'$ to the center of the incircle. Since APA' passes through the center of the incircle, $AB = AC$. The diagram is symmetric about APA' and A' is the center of BC. We have $MN \cap BC$ at the point at infinity meaning $(MN \cap BC, A'; B, C) = -1$. By the converse of lemma 9.11 on triangle PBC, segments CM, BN, PA' are concurrent.

Problem 14 (15TSTST2)

It suffices to prove that $(X_A X_B X_C)$ is the circle with diameter GH. We prove that $\angle GX_AH = 90 \Leftrightarrow \angle AX_AH = 90 \Leftrightarrow X_A \in (AXHY)$ where X, Y are bases of altitudes from C and B. Let X_A' be the second intersection of $(AXHY)$ and (AK_AL_A) other than A. We show that AX_A' is the median to get $X_A' = X_A$. Since AX_A' is also the radical axis, it is equivalent to prove that M_A has the same power with respect to both circles. Indeed, since $(B, C; K_A, L_A) = -1$, we have $M_A K_A \cdot M_A L_A = MC^2$. By Three Tangents lemma, $M_A Y$ is tangent to $(AXHY)$. Since M_A is the center of $(BCYX)$, we additionally have $M_A Y^2 = MC^2$ which finishes.

28 clubs; 6 hours

<u>Problem 15 (17SLG4)</u>

Let M_1 be the base of the altitude from I_A to AD. Let G be the reflection of D over M_1. We claim that G lies on (FDE), (QMP). The former is true by symmetry since I_A is the center of (FDE). The latter is true by Power of a Point on (AQP) and (MQP). Therefore, (FDE), (QMP), and AD are concurrent at G.

Since AE, AF are tangent to the A-excircle and ADG is collinear, we know that $FDEG$ is a harmonic quadrilateral. Therefore, the tangent to the A-excircle at D and G intersect at point W on line EF. Note that the tangent at D passes through QP meaning it's the radical axis of (AQP) and (MQP). EF is the radical axis of the A-excircle and (AFE). By radical center, we therefore have WG is the radical axis of the A-excircle and (QMP). Since WG is tangent to one of those, it must be tangent to both of those.

<u>Problem 16 (19IGOA3)</u>

Claim 1: BM, AN, XY are concurrent.

Let $M_1 = XY \cap AB$. By POP M_1 is the midpoint of AB. Note that $(XMYB)$ is a harmonic quadrilateral as the tangents at $X, Y, \overline{BM}$ concur at L. Therefore, the tangent at B, M concur on XY at M_1. We hence have $BM_1 = M_1 A = M_1 M$ meaning $\angle AMB = 90$. Analogous work gives $\angle ANB = 90$. Note that $(AMNB)$ is cyclic centered at the midpoint of AB. By radical axis, both $BM \cap AN$ and the midpoint of AB lie on XY.

With the claim proven, we let $Q = BM \cap AN \cap XY$. Projecting from B, $(XMYB)$ is harmonic then $(A, N; Q, K) = -1$. Projecting from B onto O_1O_2, $(AB \cap O_1O_2, BN \cap O_1O_2; L, K) = -1$. Doing analogous work from A gives $(AB \cap O_1O_2, AM \cap O_1O_2; L, K) = -1$. Therefore, $AM \cap O_1O_2 = BN \cap O_1O_2$ as desired.

36 clubs; 9 hours

<u>Problem 17 (18SHRG20)</u>

Let the other intersection of AN with the incircle be J. By a lemma from EGMO Chapter 4, JX is the diameter of the incircle. Let $JX \cap ZY = P$. By another lemma from EGMO Chapter 4, AP passes through M the midpoint of BC.

Since the tangents to the incircle at Y, Z intersect at A and $\overline{AJT}$ is collinear, $(ZJYT)$ is harmonic. Projecting from X onto ZY, $-1 = (ZY; PK) \stackrel{(A)}{=} (B, C; M, AK \cap BC)$. We therefore have $AK \cap BC$ at P_∞ meaning $AK \parallel BC$ as desired.

Problem 18 (05SAF4)

Note that $(DEQF)$ is a harmonic quadrilateral. Projecting from E onto AB,

$$(DE \cap AB, EQ \cap AB; F, A) = -1$$

$EQ \cap AB$ is the midpoint of AF if and only if $DE \parallel AB$ meaning $BC = AC$.

Problem 19 (05SLG6)

Let D be the point of tangency of the incircle on BC and I be the incenter. Let E, F be the other points of tangencies. Let $DI \cap AM = P$. Let $LY \cap AP = J$. We prove that M is the midpoint of PQ. By a homothety at A, this is equivalent to proving,

$$LY = LJ \Leftrightarrow \frac{LJ}{KX} = \frac{LY}{KX} \Leftrightarrow \frac{AL}{AK} = \frac{PL}{PK} \Leftrightarrow (AP; KL) = -1$$

By EGMO Ch. 4 lemma, AM, DI, EF are concurrent. Projecting from E onto the incircle,

$$\Leftrightarrow (EKFL) \text{ is harmonic}$$

This is true since the tangent at E, F and $\overline{KL}$ concur at A.

46 clubs; 11.5 hours

Problem 20 (18CSMO6)

Let E' be the E-antipode. Note that $\angle AGE = 90$ if and only if A, G, E' are collinear.

Let $E'K$ intersect the circle again at X. Note that $(FXGE')$ is a harmonic quadrilateral. Projecting from E' onto AB, noting that the tangent to the circle at E' intersects AB at B,

$$(B, K; F, A) = -1$$

Projecting from C onto EF, we are done knowing that $EF \parallel BC$.

Problem 22 (H2648084)

Let $Z = BI_A \cap AC$. Since $\angle ZBE = 90$ and $\angle CBE = \angle EBA$, we know $(ZE; AC) = -1$. Let X be the second intersection of EI_A with cyclic quad $(BICI_A)$. Projecting from I_A onto the cyclic quad, we find that $(BIXC)$ is harmonic. Therefore, BX is the symmedian of BIC.

$$\angle EI_AI = \angle XI_AI = \angle XBI = \angle CBM = \angle CBG$$

Similarly, we may get $\angle FI_AI = \angle BCG$. Adding them up, we find $\angle BGC = \angle EI_AF$ as desired.

54 clubs; 14.5 hours

Problem 24 (18HMMTG8)

Let $D = AZ \cap BY \cap CX$. Note that $\angle CXA = \angle BYC$ meaning $(AYDX)$ is cyclic. Let $E \coloneqq XX \cap YY$. However, EXY is an equilateral triangle since $\angle EXY = \angle EYX = \angle A$. If we let E' be the point on segment BC such that $BE = 5$, we also have $XE'Y$ is equilateral. Therefore, $E' = E$ as they clearly lie on the same side of XY meaning E lies on BC. Pascal's on $(XXPYYQ)$ gives Z, E, K collinear meaning K lies on BC.

Let $XY \cap BC = F$. By EGMO lemma, $(FZ; BC) = -1$. Projecting from Y onto the circle, $(AXDQ)$ is a harmonic quadrilateral. Therefore, AA, DD, QX, PY, BC all concur at K.

Using mass points, we may get $DX = \frac{9}{7}; DY = \frac{25}{7}$. Therefore,

$$|(X, Y; D, A)| = \frac{9}{25} \div \frac{5}{3} = \frac{27}{125}$$

Projecting from A onto BC,

$$|(B, C; Z, K)| = \frac{27}{125}$$

Knowing $\frac{BZ}{ZC} = \frac{9}{25}$ by Ceva's, we may get $CK = 12$. We finish by doing,

$$KQ \cdot KX = KA^2 = KB^2 + AB^2 - 2 \cdot AB \cdot KB \cdot \cos(60) = 20^2 + 8^2 - 20 \cdot 8 = 304$$

Problem 25 (15TWNQ3J6)

Let $P \coloneqq BI \cap EF$ and $Q \coloneqq CI \cap EF$. Let D be the remaining tangency point of incircle. Note that (DPQ) is the nine-point circle as $DI \cap BC, CP \cap BI, CI \cap BQ$ by Iran lemma. By an EGMO Chapter 4 config, note that $DI \cap EF \cap AM$ concur. We call the point X.

Note that $(BC; DT) = -1$ since AD, BE, CF are concurrent. Therefore, projecting from I onto EF,

$$(XT; PQ) = -1$$

Therefore, the polar of T passes through S.

Now, let $N = AI \cap EF$. Note that $(SN; AI) = -1$. Projecting from X onto BC,

$$(XS \cap BC, T; M, D) = -1$$

Therefore, the polar of T passes through $XS \cap BC$. We hence have the polar of T passes through S. EGMO lemma 9.27 suffices.

63 clubs; 17 hours